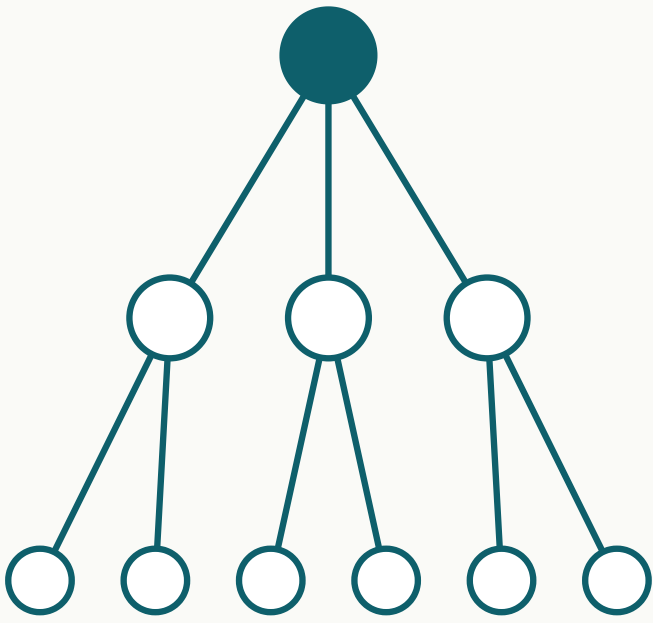


# The Bruhat–Tits Tree Code — Status and Test Plan

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## Why it matters

Fault tolerance today means surface codes: many hundreds of physical qubits per logical qubit, active syndrome measurement, millikelvin cooling. The tree code is a different geometry whose **simulated** thresholds run 4.6–75  above surface-code figures — and whether that survives a matched-noise comparison is test T1. If it does, the payoff is energy: fewer joules per correct answer.



The Bruhat–Tits tree  $T_2$  — the geometry behind the code. Every internal vertex carries a  $[[3,1,1]]$  perfect tensor.

## Verified

Thresholds computed three ways — analytical recursion, exact stabilizer simulation, Monte Carlo — and re-checked by a deposited 35-check script.

10.5281/zenodo.20109836 · 10.5281/zenodo.22038733

## Boundary

Under independent errors the *qudit* (higher-dimensional) generalization of this family sits at  $\approx 2 \times 10^{-4}$ , about 55  below surface codes. The qubit tree code's advantage claim targets correlated-failure regimes — exactly what T1 probes.

10.5281/zenodo.21046993 · 10.5281/zenodo.22025544

## Not yet

No hardware run. Tests T3–T4 are the instrumentation path.

## The code and the numbers

A holographic perfect-tensor code on a truncated Bruhat–Tits tree: a  $[[3,1,1]]$  perfect tensor sits at every internal vertex; logical information lives at the root and physical qubits at the leaves. Each block is a one-to-two-qubit isometry with distance  $d = 1$  — protection emerges from concatenation across the tree, not from any single block. Error correction is built into the geometry rather than applied by active measurement loops.

| Error model (qubit, code-capacity) | Tree code | Surface code | Ratio |
|------------------------------------|-----------|--------------|-------|
| Bit-flip                           | 50.0%     | ~10.9%       | 4.6   |
| Depolarizing                       | 75.0%     | ~1.0%        | 75    |
| Independent X+Z                    | 17.30%    | ~18.9%       | 0.92  |

Classical simulation only. Bit-flip and X+Z: i.i.d. noise, crossing criterion (the rate at which more levels stop helping) — and on X+Z the tree sits slightly below the toric code's ~18.9%. The depolarizing figure comes from the code's own structured-noise simulation; the channel is not yet published (roadmap D3), and T1 is the matched-noise comparison that settles it. Surface comparators are the standard published figures (bit-flip ~10.9% code-capacity, depolarizing ~1% circuit-level). No tree code has run on quantum hardware yet.

## The tests that would change my mind

### T1 • Matched-noise comparison

Tree vs surface under the *same* correlated noise model. Classical, days–weeks.

**Would change my mind:** the advantage falls to parity.

### T2 • Decoder + full overhead

Physical qubits plus classical decoding cost. Classical.

1. A correlated-noise benchmark — model and data — I can run T1 against.
2. A decoder reality check: does a known decoder family already cover tree-structured codes, and how does it scale?
3. Pointers on perfect tensors for  $p > 2$  (the binary  $[[3,1,1]]$  case is settled).



**Would change my mind:** the advantage disappears once decoding cost is counted.

**T3 • Random walk on a p-regular tree**

Return-probability decay. Ion / analog simulator, ~8 weeks.

**Would change my mind:** no change in the decay exponent where predicted.

**T4 • Clock-rest split**

Diagonal 0% vs non-diagonal 29–35% violation rate. Trapped ion, ~8 weeks.

**Would change my mind:** the measured split falls outside that range.

**T5 • Independent re-run**

An outside group repeats T1 with my simulator and the protocol.

**Would change my mind:** nothing to predict — this one is about reproducibility.

**4. An independent re-run of T1.**

The five tests are the takeaway: every criterion is stated with numbers, so a null result is a result. The simulator and the protocol are handed over to anyone who wants to run T1 or T5 themselves.

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**All numbers are classical simulation** — bit-flip and X+Z under i.i.d. noise (crossing criterion); the depolarizing figure is the code's structured-noise simulation (channel pending — D3). Nothing has run on hardware yet. · DOIs (verified 21 Aug 2026): 10.5281/zenodo.20109836 (architecture) · 10.5281/zenodo.22038733 (guide + verification script) · 10.5281/zenodo.21046993 (qudit QEC) · 10.5281/zenodo.22025544 (trapped-ion testbed) · 10.5281/zenodo.21964674 (QEC–Darwinism audit) · 10.5281/zenodo.21979060 (prime-valuation correction) · Full plan: [github.com/QNFO/cwi-qec-poster-2026](https://github.com/QNFO/cwi-qec-poster-2026) → [docs/tree-code-roadmap.md](#)